THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS UNIT TEST SET A

Chapters 5–8: **Circles, Measuring Space: Perimeter and Area, The Mathematics of Maybe: Introduction to Probability, Predicting What Comes Next: Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) — 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. A chord of length 16 cm is at a distance of 6 cm from the centre of a circle. The radius of the circle is: 1

(a) 14 cm **(b)** 16 cm **(c)** 11 cm **(d)** 10 cm

2. In a circle of radius 15 cm, a chord is 9 cm from the centre. The length of the chord is: 1

(a) 22 cm **(b)** 12 cm **(c)** 26 cm **(d)** 24 cm

3. Points lying on the same circle are called: 1

(a) concurrent points **(b)** coplanar points
(c) concyclic points **(d)** collinear points

4. By Brahmagupta's formula, the area of a cyclic quadrilateral with sides $5, 5, 5, 5$ is: 1

(a) 23 sq. units **(b)** 29 sq. units **(c)** 25 sq. units **(d)** 50 sq. units

5. The perimeter of a circle is 66 cm. Taking $\pi = \frac{22}{7}$, its radius is: 1

(a) 10.5 cm **(b)** 21 cm **(c)** 5.25 cm **(d)** 11.5 cm

6. Taking $\pi = \frac{22}{7}$, the circumference of a circle of radius 7 cm is: 1

(a) 44 cm **(b)** 88 cm **(c)** 22 cm **(d)** 154 cm

7. A letter is picked at random from the word ‘PROBABILITY’. The probability of picking a vowel is: 1

(a) $\frac{5}{11}$ **(b)** $\frac{4}{11}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{7}{11}$

8. Believing that tails is ‘due’ after many heads is called: 1

(a) the Gambler's Fallacy **(b)** relative frequency
(c) sampling **(d)** the Law of Large Numbers

9. A sample shows a probability of $\frac{2}{5}$ that a student likes cricket. In a school of 1500 students the expected number who like cricket is: 1

(a) 300 **(b)** 700 **(c)** 750 **(d)** 600

10. The sequence $1, 4, 9, 16, \ldots$ is: 1

(a) neither an AP nor a GP **(b)** a GP
(c) an AP **(d)** both an AP and a GP

11. The 9th term of the AP $11, 8, 5, \ldots$ is: 1

(a) −12 **(b)** −13 **(c)** −14 **(d)** −26

12. For the sequence $t_n = 3n - 4$, the value of t_{12} is: 1

(a) 31 **(b)** 64 **(c)** 32 **(d)** 33

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

13. PQRS is cyclic with $\angle P = (2x + 10)^\circ$ and $\angle R = (3x - 20)^\circ$. Find x. 2

14. Find the perimeter of a semicircular plate of radius 7 cm (including the diameter). ($\pi = \frac{22}{7}$) 2

15. The perimeters of two circles are in the ratio 5 : 4. What is the ratio of their radii? 2

16. A fair die is rolled. Find the probability of getting a prime number. 2

17. Let $T_1 = 1, T_2 = 2, T_3 = 4$ and $T_n = T_{n-1} + T_{n-2} + T_{n-3}$. Find T_4, T_5, T_6. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. Two parallel chords of length 6 cm and 8 cm are in a circle of radius 5 cm. Find the distance between them if they are on opposite sides of the centre. 3

19. Find the length of the altitude on the shortest side of a triangle with sides 7 cm, 15 cm and 20 cm. 3

20. In a sample of 100 students, 35 like cricket. Estimate how many of 1200 students like cricket, and explain why a larger, more varied sample gives a better estimate. 3

21. Harish starts at ₹5,00,000 a year and gets a raise of ₹20,000 each year. After how many raises does his salary reach ₹6,00,000? 3

22. How many two-digit numbers are divisible by 3? Find their sum. 3

OR

For the sequence $1, 4, 7, 10, \ldots$ write the explicit rule and the recursive rule. 3

23. A cyclic quadrilateral has sides 5, 5, 12, 12 units in order. Find its area. 3

OR

AB is a diameter of a circle, C is on the circle and $\angle CAB = 35^\circ$. Find $\angle ABC$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. **Case study:** A circular garden of radius 14 m has a 7 m wide path around it. ($\pi = \frac{22}{7}$) 5

(i) Find the area of the garden.
(ii) Find the outer radius and the area of the garden together with the path.
(iii) Find the area of the path.
(iv) Find the cost of paving it at ₹50 per m^2.

OR

Case study: A 400 m track has two straights of 84.39 m each and two semicircular ends. The innermost semicircle has radius 36.5 m and each lane is 1.22 m wide. A runner keeps 0.3 m from the inner edge. 5

(i) Find her radius on the bends.
(ii) Find the distance she runs in one circuit ($\pi = 3.1416$).
(iii) Find the stagger needed between adjacent lanes.
(iv) Why must the stagger be the same for every pair of adjacent lanes?

25. **Case study:** A ball is dropped from 24 ft and rebounds to $\frac{3}{4}$ of its previous height each time. 5

(i) Find the heights after the first five bounces.
(ii) Write the nth term.
(iii) After how many bounces does the height first fall below 3 ft?

OR

Case study: Harish's starting salary is ₹5,00,000 and he gets an increment of ₹20,000 each year. 5

(i) Write his salary in the first three years.
(ii) Find the salary in the 8th year.
(iii) In which year will it be ₹8,00,000?
(iv) Find his total earnings in 10 years.

END OF QUESTION PAPER — **Total: 50 Marks**